

Specialization:

Healthcare Analytics

Health Focus:

Healthcare & Oncology Research

Skill Outcome:

- Advanced SQL Querying & Joins
- Healthcare Data Analysis

Dermalife Oncology & Research Institute (DORI)

Skin Cancer Risk Intelligence & Patient Diagnosis Optimization

A HEALTHCARE DATA ANALYTICS CASE STUDY

Project Learning Opportunities:

- Analyze real-world healthcare datasets using SQL
- Identify high-risk patient and diagnosis patterns
- Perform healthcare trend and risk analysis
- Generate actionable business and healthcare insights
- Develop data-driven recommendations for decision-making

Tools and Technology Used:

Company Introduction

Dermalife Oncology & Research Institute is a specialized healthcare organization dedicated to combating skin cancer through clinical research, advanced diagnostics, and population health analytics.

The institute manages thousands of patient records across multiple healthcare regions and continuously monitors patient symptoms, lesion characteristics, hereditary risk factors, and environmental conditions to improve diagnostic accuracy and treatment outcomes.

DORI's mission is to leverage healthcare data to:

- Reduce skin cancer mortality rates
- Improve early-stage diagnosis success
- Deliver personalized patient risk assessments
- Support evidence-based medical decisions
- Increase awareness of environmental and behavioral health risks

As part of its digital transformation initiative, DORI launched a healthcare analytics project to uncover actionable insights from its patient database using SQL and business intelligence techniques.



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The Health Problem

DORI has experienced a **significant rise in late-stage skin cancer diagnoses, high-risk lesion progression, and recurring patient complications, particularly among elderly and underserved populations.** Healthcare administrators also suspect that **environmental conditions, hereditary history, smoking, alcohol consumption, and poor sanitation access** are **major contributors to the increasing skin cancer risk.**

The Problem

Purpose: Rising Skin Cancer Cases & Delayed Diagnosis .

Despite having **large volumes** of patient data, **the organization lacked a structured analytical framework to identify:**

- Patients most at risk
- Regions requiring urgent intervention
- High-risk symptoms linked to malignancy
- Lifestyle factors worsening patient outcomes
- Early warning indicators for dangerous lesion growth

This created the need for a **robust healthcare analytics initiative powered by SQL and data-driven insights.**

Your Role

You have been engaged as a **Healthcare Data Analyst Consultant** to **analyze patient, diagnosis, and environmental health data using SQL and analytics techniques.** Your role is to identify high-risk patients, uncover diagnosis trends, and provide data-driven recommendations that improve early detection and preventive healthcare strategies.



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Email structure



Dear Healthcare Analytics Team,

Dermalife Oncology & Research Institute (DORI) is seeking analytical support to investigate rising skin cancer cases, delayed diagnoses, and high-risk patient trends across our healthcare network.

Using our patient and lesion datasets, we require a SQL-driven analysis to identify key risk factors, diagnosis patterns, regional disparities, and opportunities for early detection improvement.

The project deliverables should include analytical queries, strategic insights, and actionable healthcare recommendations.

Kind regards,

Dr. Amanda Reynolds

Director of Clinical Research

Dermalife Oncology & Research Institute (DORI)



Rationale for the Project

This Project should....

Early Risk Detection



This project will help identify high-risk patients early to improve diagnosis speed and reduce severe skin cancer cases.

Data-Driven Healthcare Decisions



This project will provide actionable insights that support better clinical decisions and preventive healthcare strategies.

Regional & Lifestyle Risk Monitoring



This project will uncover how environmental, hereditary, and lifestyle factors contribute to skin cancer risks across different populations.



Project Initiative

By completing this project, you will be able to:



Patient Risk Identification

Identify high-risk patients using demographic, lifestyle, and medical history data.



Diagnosis Trend Analysis

Analyze lesion symptoms and diagnosis patterns to improve early skin cancer detection.



Environmental Risk Assessment

Evaluate how environmental and regional factors influence skin cancer prevalence and severity.



Preventive Healthcare Strategy

Generate data-driven recommendations that support preventive healthcare planning and patient care improvement.

Project Workflow



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Step 1



Import and restore the healthcare database into PostgreSQL for analysis and query execution.

Step 2



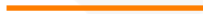
Explore patient demographics, lesion characteristics, symptoms, and diagnosis distributions.

Step 3



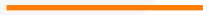
Use SQL queries to identify high-risk patients, diagnosis trends, and environmental risk factors.

Step 4



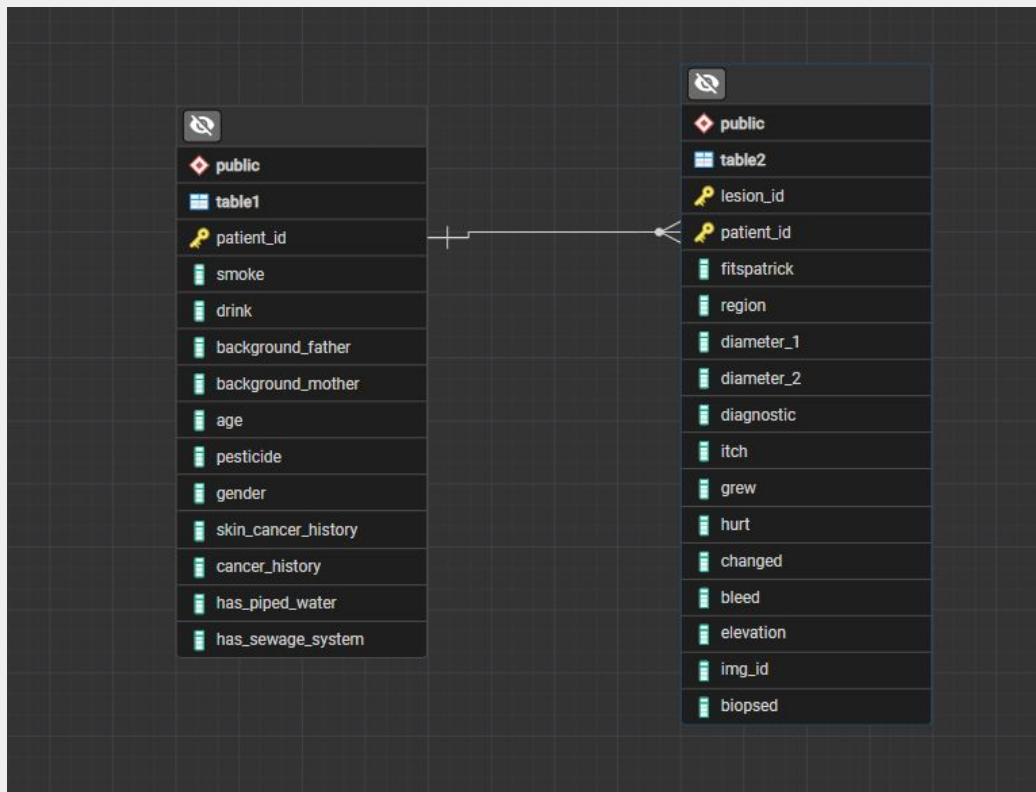
Generate actionable insights and strategic healthcare recommendations to improve early detection and preventive care.

Step 5



Create a presentation summarizing key insights, analytical findings, and healthcare recommendations from the project.

Entity Relationship diagram



Click To [Download the Dataset](#)

Data Dictionary- Table 1



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Column Header	Description	
patient_id	Unique identifier assigned to each patient	VARCHAR
smoke	Indicates whether the patient smokes	BOOLEAN
drink	Indicates whether the patient consumes alcohol	BOOLEAN
background_father	Patient's paternal ancestry/background	VARCHAR
background_mother	Patient's maternal ancestry/background	VARCHAR
age	Patient age	INTEGER
pesticide	Indicates exposure to pesticides	BOOLEAN
gender	Patient gender	VARCHAR

Data Dictionary- Table 1



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Column Header	Description	
skin_cancer_history	Indicates previous skin cancer history	BOOLEAN
cancer_history	Indicates previous cancer history	BOOLEAN
has_piped_water	Indicates access to piped water	BOOLEAN
has_sewage_system	Indicates access to sewage systems	BOOLEAN

Data Dictionary– Table 2



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Column Header	Description	
patient_id	Unique identifier assigned to each patient	VARCHAR
lesion_id	Unique identifier assigned to each lesion	INTEGER
fitzpatrick	Skin type classification based on sensitivity to UV exposure	INTEGER
region	Body region where the lesion is located	VARCHAR
diameter_1	Primary lesion diameter measurement	NUMERIC
diameter_2	Secondary lesion diameter measurement	NUMERIC
diagnostic	Final diagnosis category for the lesion	VARCHAR
itch	Indicates whether the lesion causes itching	BOOLEAN

Data Dictionary– Table 2



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grew	Indicates whether the lesion has increased in size	BOOLEAN
hurt	Indicates whether the lesion is painful	BOOLEAN
changed	Indicates whether the lesion appearance changed	BOOLEAN
bleed	Indicates whether the lesion bleeds	BOOLEAN
elevation	Indicates whether the lesion is elevated	BOOLEAN
img_id	Image identifier linked to lesion records	VARCHAR
biopsed	Indicates whether the lesion was biopsied for further medical examination and confirmation	BOOLEAN

→ Key Analytical Questions



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Task 1: Patient Demographic Risk Analysis

Goal: Understand which demographic groups are most affected by skin cancer conditions. .

1. Which age group has the highest number of skin cancer diagnoses?
2. What is the distribution of diagnoses between male and female patients?
3. Which body regions record the highest number of patients with malignant diagnoses?
4. How many patients have a previous history of skin cancer?

→ Key Analytical Questions



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Task 2: Lesion Growth & Diagnosis Analysis

Goal: Analyze lesion characteristics and identify indicators of dangerous skin conditions.

1. Which diagnosis category appears most frequently?
2. How many lesions were reported as growing over time?
3. Which symptoms are most commonly associated with lesions?
4. How many lesions were biopsied before diagnosis confirmation?
5. Which diagnosis type has the highest average lesion diameter?

→ Key Analytical Questions



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Task 3: Environmental Healthcare Analysis

Goal: Understand how environmental conditions influence diagnosis patterns.

1. Which body region has the highest number of diagnosed cases?
2. How many patients lack access to piped water?
3. How many patients do not have access to sewage systems?
4. Which body regions report the highest number of biopsied lesions?
5. Is there a relationship between poor sanitation access and severe diagnosis outcomes?

→ Key Analytical Questions



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Task 4: Lifestyle & Behavioral Risk Analysis

Goal: Evaluate how lifestyle habits contribute to skin cancer risks and lesion severity.

1. How many patients are smokers?
2. How many patients consume alcohol regularly?
3. Which diagnosis types are most common among smokers?
4. What percentage of smokers also consume alcohol?
5. Are patients who both smoke and drink more likely to develop malignant conditions?
6. Which lifestyle factor has the strongest relationship with severe diagnosis outcomes?

Download the database

[HERE](#)

Dive into the project with your POD members

Please note

To complete the capstone, you must submit:

- 1) The SQL Script: A .sql file containing all queries used to answer the questions above.
- 2) The PDF Presentation: A slide including a Title Slide, Business Overview, Business problem, Analysis, Insights, and a Final Recommendation.

Don't forget to add it to your portfolio and post on LinkedIn and tag [@10Alytics](#)

and use these

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